Bal Bharati Hackathon 3.0

Idea Submission

NIVRA: Before Impact

AI-assisted observation. Human-led decisions.

Category: Cat2

Problem Statement Title	CCTV can miss warning signs before an incident escalates
Theme	T3 | AI for Good
Team Name	Team Damage Control
Team Leader	Harsh Jain
Unit Name	Bal Bharati Public School, Pitampura
Team Size	5 students (excluding mentor)

Your Idea

IDEA / SOLUTION :



THE PROBLEM

- CCTV records incidents after they happen.
- Warning signs are easy to miss in live feeds.
- Operators need useful context, not a noisy alarm.

Problem Resolution :



OUR ANSWER: NIVRA

- Watches movement over time.
- Flags explainable warning patterns.
- Keeps a human in control of every response.

Images are generated concepts. The multi-device response network is proposed, not implemented.

Your Idea | Visible signals

WHAT NIVRA LOOKS FOR



Visible warning signals

- Rapidly closing distance
- Repeated directed arm movements
- Crowd convergence or sudden changes

WHAT THE OPERATOR DOES

01 / SEE

Tracks people and objects temporarily.

02 / CHECK

Compares movement with simple rules.

03 / REVIEW

Shows the alert reason; a person decides what happens next.

Experimental motion rules; sports, play, occlusion and camera angle can produce errors.

TECHNICAL APPROACH

Inputs

Camera, local video or scripted scenario.

Two Web Workers

One for pose, one for objects. Local MediaPipe + WASM.

Runtime

GPU for real-time pose when supported. CPU for review scans and fallback.

BROWSER ARCHITECTURE



Frames remain in the browser

No frame uploads or recordings. Tracks and review logs are session-only. No application backend is required.

TECHNICAL APPROACH | Demo

CURRENT LAB AND NEXT STEPS

Try the current lab

1. Choose camera, video or test scenario.
2. Inspect skeletons and review reasons.
3. Acknowledge alerts; pause, replay or stop.

211 tests passed

Lint and production build also passed during local verification.



Proposed next: connected response

Multi-camera coordination, an 8-second context replay and responder dispatch remain future work.

Current app: 8-second alert cooldown, not an 8-second footage buffer. Mockup is conceptual.

Feasibility and Viability

FEASIBILITY OF THE IDEA

- Technical: local model bundles, configurable Full/Lite pose and CPU recovery.

Operational: familiar camera/video controls; a person reviews each alert.

Resources: current Chrome, laptop or phone; Node 24 for local development.

POTENTIAL CHALLENGES & RISKS

- 01 Sports and play can trigger reviews.

02 Occlusion or weak devices reduce coverage.

03 School video needs permission and privacy controls.

STRATEGIES TO OVERCOME THEM

- 01 Require human review; test positive and quiet clips.

02 Use Lite / CPU fallback; show visibility and runtime state.

03 Keep frames local; obtain school approval before trials.

Impact and Benefits

Social Impact	Economic Impact	Environmental Impact
Earlier attention to visible changes, with the final decision left to staff.	Reuse existing devices where practical. No measured savings or pricing claim.	Potential hardware reuse. No measured environmental benefit.

BENEFITS OF THE SOLUTION

- End users: clear visual alerts, reasons and local acknowledgement.

Institutions: a basis for testing a supervised safety workflow.

Scalability: validate one location before multi-camera or responder integration.

Impact and Benefits | Why it matters

4.4 million	1.25 million	~12,000 / day
Global injury-related deaths in 2019, including unintentional injuries.	Violence-related injury deaths in 2019, including self-harm and conflict.	Approximate daily equivalent of 4.4 million. Not a count of public fights.

WHAT THESE FIGURES MEAN FOR NIVRA

- The scale of harm supports investigating better observation and response.

These broad global totals are not NIVRA's addressable market or preventable cases.

Product value must be demonstrated through supervised, representative tests.

Source: WHO, Preventing injuries and violence (2022), 2019 estimates.

Research and References

EVIDENCE AND SOURCES

NIVRA repository
Implementation, test fixtures and limitations.

WHO | Injuries and violence
Broad global burden; not product impact.

MediaPipe Pose Landmarker
Landmark API and model documentation.

MoE school safety / DPDP Act
Policy context and privacy review before trials.

Axis Object Analytics
Existing event-analytics category for comparison.

Technology Stack

IMPLEMENTED IN THE REPOSITORY

React 19 + TypeScript + Vite
Browser UI and production build.

MediaPipe Tasks Vision 1.0.1
Pose Landmarker Full / Lite, float16.

EfficientDet-Lite0
Optional object analysis on CPU.

Web Workers + WASM + Canvas
Two workers; GPU attempt / CPU fallback.

Vitest + Playwright + Oxlint
Unit tests, browser tests and lint tools.

Data: bundled exercise fixture + scripted observations; representative safety evaluation remains to be done.

Team Member Details

Student's Details

SL	Student ID	Name	Class	DOB	Email	Phone Number	Gender
1	223564	Harsh Jain	12C	23/04/2009	223564@pp.balbharati.org	9810141505	Male
2	218461	Tanishq Aggarwal	12C	09/05/2009	218461@pp.balbharati.org	9810398813	Male
3	201362	Keshav Garg	12D	11/06/2009	201362@pp.balbharati.org	7011900075	Male
4	213266	Reyansh Garg	12D	02/07/2009	213266@pp.balbharati.org	8595774056	Male
5	213283	Asmi Sharma	12D	28/03/2010	213283@pp.balbharati.org	8595612803	Female

Mentor Details

SL	Name	UID	Address	DOB	Email	Phone Number	Gender
1							